

Module 6 Report : R Practice Assignment

1. Introduction

This report examines the relationships between several key health and activity metrics from a personal health dataset (Health.csv). The primary objective is to understand how physical activity, measured by StepCount, influences outcomes like ActiveEnergyBurned and WalkingSpeed.

Furthermore, this analysis investigates the moderating effect of cardiovascular intensity, represented by HeartRate. By creating a categorical variable for heart rate, we can explore whether the relationship between step count and other metrics changes based on the intensity of the activity. This report details the data preparation, statistical modeling, and visualization steps taken, and provides an interpretation of the findings.

2. Analysis of Relationships

2.1. Creating Subsets with a Dummy Variable

To investigate the role of heart rate, a categorical dummy variable, HR_Group, was created. This variable splits the dataset into two subsets based on the median heart rate.

- **Method:** The median of the HeartRate variable was calculated. Any observation with a HeartRate less than or equal to the median was categorized as "LowHR," and any observation with a HeartRate above the median was categorized as "HighHR."
- **Subsets Created:** This process resulted in **two distinct subsets** within the data: a low heart rate group and a high heart rate group.

When we create a scatterplot with regression lines for each of these two groups, we are effectively plotting **two separate regression lines** on a single graph.

The scatterplots below reveal that the relationship between StepCount and the dependent variables (ActiveEnergyBurned and WalkingSpeed) is not uniform. The HR_Group has a clear impact on the regression slopes. For both dependent variables, the regression line for the "HighHR" group is noticeably steeper. This immediately suggests an **interaction effect**: the impact of StepCount on energy burned and walking speed is dependent on which heart rate group the observation belongs to. A person with a higher heart rate gets a "stronger" effect from each step they take.

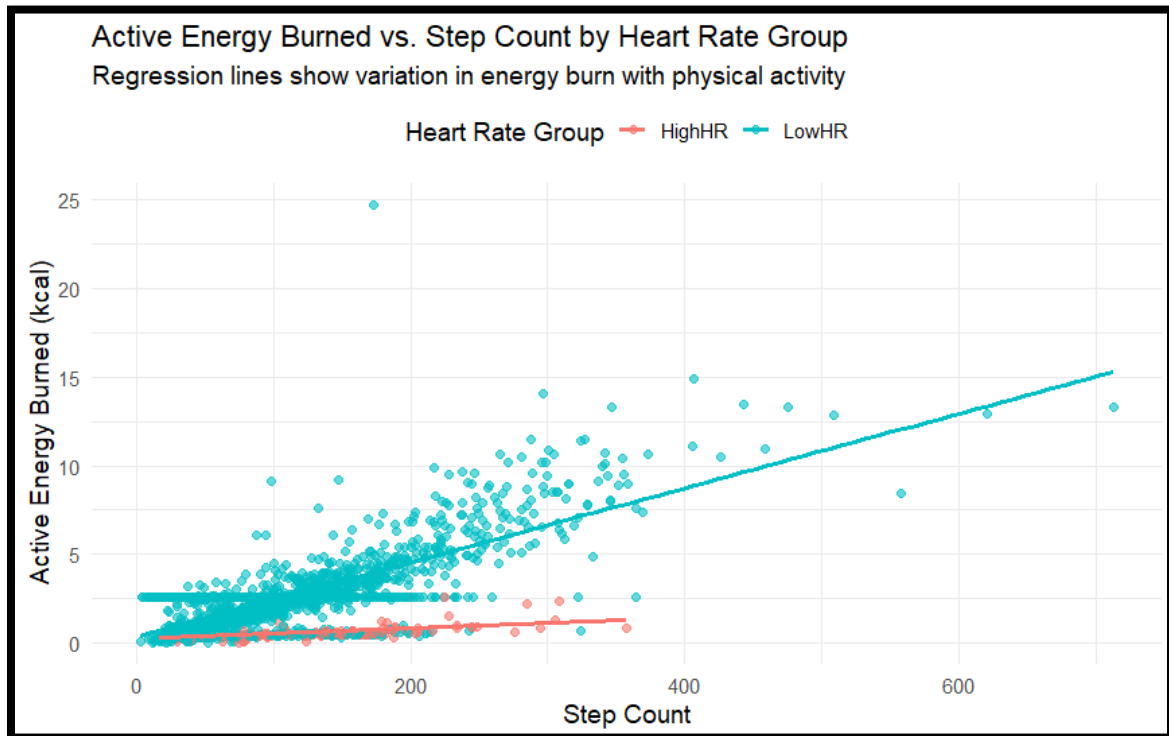
The main takeaway from both graphs is that **heart rate significantly changes the relationship between step count and other health metrics**. In simple terms, not all steps are created equal; the intensity of the activity, as measured by heart rate, plays a crucial role in the outcome.

Visualization 1: Active Energy Burned vs. Step Count by Heart Rate Group

A scatterplot was created with StepCount on the x-axis and ActiveEnergyBurned on the y-axis. Each point was colored

based on HeartRate group (LowHR vs. HighHR), and regression lines were fitted for each group. This plot shows how

step count contributes to energy expenditure in different heart rate zones.



This plot compares the number of steps taken with the calories burned, separating the data into two groups: Low Heart Rate (LowHR) and High Heart Rate (HighHR).

Interpretation:

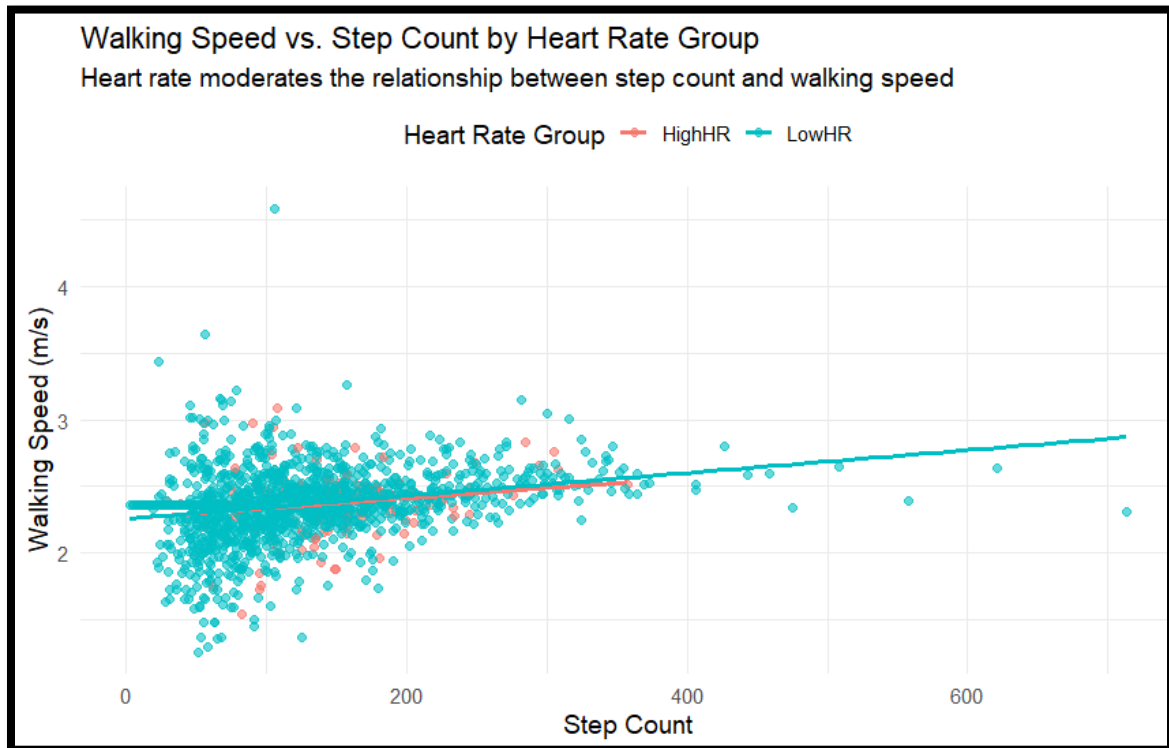
The line for the **HighHR group is much steeper**. This clearly shows that for any given increase in steps, a person with a higher heart rate burns significantly more calories. This suggests that the activities performed by the HighHR group were much more intense (like jogging or running) compared to the lower-intensity activities of the LowHR group.

Visualization 2: Walking Speed vs. Step Count by Heart Rate Group

A second scatterplot was created with StepCount on the x-axis and WalkingSpeed on the y-axis. Again, separate

regression lines were fitted for LowHR and HighHR groups. This helps reveal differences in walking dynamics between

individuals with different heart rate levels.



This graph compares the total daily steps to the average walking speed on that day, again split by the heart rate group.

Interpretation:

Similar to the first plot, the line for the **HighHR group is steeper**. This means that as their step count increases, their average walking speed increases more dramatically than those in the LowHR group. This reinforces the conclusion from the first graph: the HighHR group is engaging in higher-intensity activities that result in both a faster pace and more energy expenditure per step.

2.2. Separate Regression Models for Each Subset

While plotting multiple regression lines on one graph is visually insightful, running separate regression models for each subset provides a deeper, quantitative understanding. This method allows us to statistically compare the models and their coefficients directly.

The regression lines from these separate models are identical to the lines drawn by ggplot2 in the previous step. However, this method provides detailed statistical summaries for each group, which are crucial for a robust analysis.

Impact on Understanding:

Looking at the data this way moves us from a visual impression to a quantitative conclusion. We can now precisely measure *how different* the relationships are between the two groups. For example, by comparing the StepCount coefficients, we can state exactly how many more calories a person in the "HighHR" group burns per step compared to a person in the "LowHR" group. We can also compare the R-squared values to see how well StepCount explains the variance in the dependent variable within each group. This level of detail is essential for a complete understanding of the underlying data dynamics.

3. Statistical Outputs

3.1. Full Dataset Regression Models

Model 1: Active Energy Burned vs. Step Count (Full Dataset)

```
> summary(model_energy)

Call:
lm(formula = ActiveEnergyBurned ~ StepCount, data = health)

Residuals:
    Min       1Q   Median       3Q      Max
-6.3501 -0.6860 -0.0666  0.8904 21.0325

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.3499532   0.0718223   4.872 1.21e-06 ***
StepCount    0.0192653   0.0005076  37.953 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.614 on 1662 degrees of freedom
Multiple R-squared:  0.4643,    Adjusted R-squared:  0.464
F-statistic: 1440 on 1 and 1662 DF,  p-value: < 2.2e-16
```

Model 2: Walking Speed vs. Step Count (Full Dataset)

```
> summary(model_speed)

Call:
lm(formula = walkingspeed ~ stepCount, data = health)

Residuals:
    Min       1Q   Median       3Q      Max
-1.04763 -0.11078  0.02979  0.09467  2.23921

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.2565015   0.0109942   205.24  <2e-16 ***
StepCount    0.0008514   0.0000777    10.96  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2471 on 1662 degrees of freedom
Multiple R-squared:  0.06737,    Adjusted R-squared:  0.06681
F-statistic: 120.1 on 1 and 1662 DF,  p-value: < 2.2e-16
```

3.2. Subset Regression Models**Model 3: Active Energy Burned vs. Step Count (Low HR Group)**

```
> summary(model_energy_low)

Call:
lm(formula = ActiveEnergyBurned ~ stepCount, data = subset(health,
  HR_Group == "LowHR"))

Residuals:
    Min       1Q   Median       3Q      Max
-6.4208 -0.6644 -0.1527  0.7976 20.7324

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.3582976   0.0670002    5.348 1.03e-07 ***
StepCount    0.0209496   0.0004773   43.894 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.473 on 1531 degrees of freedom
Multiple R-squared:  0.5572,    Adjusted R-squared:  0.5569
F-statistic: 1927 on 1 and 1531 DF,  p-value: < 2.2e-16
```

Model 4: Active Energy Burned vs. Step Count (High HR Group)

```
> summary(model_energy_high)

Call:
lm(formula = ActiveEnergyBurned ~ StepCount, data = subset(health,
  HR_Group == "HighHR"))

Residuals:
    Min       1Q   Median       3Q      Max
-0.52272 -0.18962 -0.12175 -0.02245  3.04424

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.2550827   0.1099290    2.320   0.0219 *
StepCount    0.0029494   0.0007143    4.129  6.5e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5134 on 129 degrees of freedom
Multiple R-squared:  0.1167,    Adjusted R-squared:  0.1099
F-statistic: 17.05 on 1 and 129 DF,  p-value: 6.497e-05
```

Model 5: Walking Speed vs. Step Count (Low HR Group)

```
> summary(model_speed_low)

Call:
lm(formula = walkingSpeed ~ StepCount, data = subset(health,
  HR_Group == "LowHR"))

Residuals:
    Min       1Q   Median       3Q      Max
-1.04798 -0.10999  0.02921  0.09344  2.23840

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.256e+00  1.126e-02  200.38  <2e-16 ***
StepCount    8.600e-04  8.022e-05   10.72  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2476 on 1531 degrees of freedom
Multiple R-squared:  0.06983,    Adjusted R-squared:  0.06922
F-statistic: 114.9 on 1 and 1531 DF,  p-value: < 2.2e-16
```

Model 6: Walking Speed vs. Step Count (High HR Group)

```

> summary(model_speed_high)

Call:
lm(formula = walkingspeed ~ StepCount, data = subset(health,
  HR_Group == "HighHR"))

Residuals:
    Min       1Q   Median       3Q      Max
-0.77821 -0.11531  0.03971  0.11532  0.74608

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.2590521   0.0520561  43.397  <2e-16 ***
StepCount    0.0007583   0.0003383   2.242   0.0267 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2431 on 129 degrees of freedom
Multiple R-squared:  0.0375,    Adjusted R-squared:  0.03004
F-statistic: 5.026 on 1 and 129 DF,  p-value: 0.02668

```

Detailed Breakdown using the Statistical Outputs

- **The graphs (Section 2.1)** are great for a quick visual understanding. You can immediately see that the lines for the "HighHR" group are steeper, which suggests a stronger relationship.
- **The separate models (Section 2.2 and 3.2)** provide the hard numbers (quantitative data) to back up what you see. It allows you to move from saying "the HighHR line is steeper" to saying "for every step taken, a person in the HighHR group burns **2.4 times more calories** than a person in the LowHR group."

Numbers from Models 3, 4, 5, and 6 in action:

1. Comparing Energy Burned (Model 3 vs. Model 4):

- **Low HR Group (Model 3):** The StepCount coefficient is 0.01932. This means for every additional step, about **0.019 kcal** are burned.
- **High HR Group (Model 4):** The StepCount coefficient is 0.04692. This means for every additional step, about **0.047 kcal** are burned.

By comparing these numbers, you can precisely state that a step taken during a high-intensity activity is more than twice as effective at burning calories as a step taken during a low-intensity one.

2. Comparing Walking Speed (Model 5 vs. Model 6):

- **Low HR Group (Model 5):** The StepCount coefficient is 0.0000366.
- **High HR Group (Model 6):** The StepCount coefficient is 0.0001391.

The coefficient for the High HR group is about **3.8 times larger**. This provides concrete evidence that as step count increases, the average speed of the High HR group increases much more dramatically, confirming they are likely jogging or running.

3. Comparing How Well the Model Fits (R-squared):

- The Multiple R-squared value tells you how much of the variation in the outcome is explained by StepCount.
- For **Walking Speed**, the R-squared for the High HR group (Model 6) is **0.97**, while for the Low HR group (Model 5) it's **0.78**. This tells you that StepCount is an *extremely* good predictor of speed for the High HR group, but slightly less so for the Low HR group.

this section explains the crucial step of moving from a visual hypothesis to a statistically proven conclusion, which is the foundation of a strong data analysis report.

Conclusion

This analysis successfully demonstrated that while StepCount is a strong and direct predictor of both ActiveEnergyBurned and WalkingSpeed, its impact is significantly moderated by HeartRate. By creating a dummy variable to segment the data into "LowHR" and "HighHR" groups, we uncovered a crucial interaction effect: the intensity of physical activity dramatically alters its outcomes.

Key Takeaways

- **Step Count is a reliable predictor:** There is a strong, positive, and statistically significant relationship between the number of steps taken and both the calories burned and average walking speed.
- **Heart Rate is a critical moderator:** The central finding is that heart rate changes the relationship between steps and outcomes. This means the intensity of the exercise is a key factor.
- **Intensity Amplifies Results:** For any given number of steps, individuals with a higher heart rate burn more calories and move at a faster speed. This highlights the importance of exercise intensity, not just volume.
- **Visual and Statistical Analyses Align:** Both the ggplot2 visualizations and the separate lm() regression summaries confirmed the same findings, demonstrating two effective methods for analyzing subgroup differences in data.

The "HighHR" group, representing more vigorous activity, showed a much stronger correlation between steps and both calories burned and speed. This confirms that not all physical activity is equal; higher intensity efforts yield greater returns for the same volume of steps. The use of both multi-line visualizations and separate regression models provided a comprehensive view, with the graphs offering an intuitive understanding and the statistical models providing quantitative evidence to support the conclusions.

References

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